Field-cooperative neural topic modeling of civil engineering literature: missing-abstract inference

Abstract

Engineering libraries and design offices hold far more bibliographic records than any team can read one by one, and a large share of these records circulate with titles only. We consider a practical form of this problem: organising civil engineering title–abstract records into interpretable topics, and still assigning topics when the abstract is missing. **Contribution to artificial intelligence:** we present FC-NTM, a field-cooperative neural topic model in which each field of a record is encoded by its own conditional variational stream while one linear topic decoder is shared across the streams; cross-field reconstruction, a consistency term, field-level word masking and whole-field dropout allow a single trained model to serve complete records, partially observed records and title-only records, and a unified topic–word read-out protocol under one evaluation vocabulary separates read-out effects from model effects in comparative evaluation. **Engineering application:** the model is evaluated on 8,000 civil engineering title–abstract records from OpenAlex. FC-NTM achieves the highest topic coherence among the ten evaluated methods, and a ten-seed paired comparison against the biterm topic model favours FC-NTM in all ten seeds, with coherence gains of +0.049 (class term-frequency–inverse-document-frequency read-out) and +0.061 (decoder read-out). Title-only routing reaches 0.66 accuracy against automatically assigned OpenAlex subject labels on the covered subset, supporting browsing and screening of engineering records. Occlusion experiments quantify inference stability under missing-field conditions on the civil corpus, GitHub issues and AG News, and identify the corpus conditions under which field cooperation produces consistent gains.

Keywords: Neural topic model, field cooperation, neural variational inference, missing data, civil engineering literature, text mining

1 Introduction

Paper and report collections in engineering libraries and design offices grow faster than anyone can read them, and in many catalogues the abstract is simply absent — the title is all that circulates. If the records could still be sorted into interpretable topics with titles alone, staff could browse and screen a collection cheaply, leaving the fine cataloguing to experts. Title–abstract records are one instance of a pattern that recurs across engineering documents: a short summary-like field sits next to a longer

contextual field, as in a clause and the contract around it, or a report headline and its body. And the two fields rarely say the same thing in the same way. Bodies carry context and causes; summaries carry objects and actions.

Topic models reduce an unlabelled collection to a set of readable topic structures, and domain-specific text mining leans on them heavily — financial news García-Méndez et al. (2023), software repositories, social media. Latent topics have also been coupled with downstream neural encoders for large-scale text classification Qin et al. (2024). The model families themselves have shifted over the past decade, from probabilistic graphical models to neural variational ones. LDA Blei et al. (2003) is the classical bag-of-words Bayesian model. GSDMM targets short texts with a collapsed Gibbs Dirichlet–multinomial mixture Yin and Wang (2014). With AVITM, autoencoding variational inference entered the field and produced ProdLDA, whose linear topic–word decoder is easy to read Srivastava and Sutton (2017). NVDM applied variational auto-encoders to text early on Miao et al. (2016); CTM and CombinedTM added contextual sentence embeddings to bag-of-words counts Bianchi et al. (2021); ETM placed topics in a word-embedding space Dieng et al. (2020). BTM, the biterm topic model, keeps representing each text through word pairs and is still a baseline to beat on short texts Yan et al. (2013). Surveys treat structured and short-text corpora as open problems for neural topic models Wu et al. (2024).

Concatenate the fields into one bag of words, as most neural topic models implicitly do, and the internal field structure of a document is gone. A separate literature comes closer: multi-view and multi-source learning across heterogeneous observations Kurle et al. (2019); Zhang et al. (2017); Pergola et al. (2021); Liu et al. (2022), structured topic models for reviews that encode user–item context and sentence-level aspects Esmaeili et al. (2019), and structural topic models with document-level covariates Roberts et al. (2019). On the short-text side, DV-BTM optimises two views jointly Liu et al. (2022), CRNTM reinforces topic inference with context Feng et al. (2022), and DVI-DTM learns dual-view representations for dynamic topics Liu et al. (2026). In all of these the views carry known, externally given roles — source-specific observations Kurle et al. (2019), opinions and plots in reviews Pergola et al. (2021), aspects in review models Esmaeili et al. (2019). Here the roles are not given. The two fields of the *same* record are modelled by separate variational streams, and their relation comes from the record schema rather than from an extra latent variable. Separate encoders, posterior alignment and missing-view inference are all established; what this study adds is the field-cooperative configuration itself, together with the read-out protocol that measures what each field contributes.

Engineering informatics, meanwhile, has built up a text-mining practice of its own: safety reports content-analysed and classified Tixier et al. (2016); Zhang et al. (2019), regulations and contracts processed automatically down to contractual risk clauses Salama and El-Gohary (2016); Zhou and El-Gohary (2016); Moon et al. (2022); Pham and Han (2023), and dedicated surveys of natural language processing for construction Wu et al. (2022); Shamshiri et al. (2024). Topic models have joined in — DUET for physical events in urban social media Wang and Taylor (2019), a mapping of machine-learning applications in buildings Zhou et al. (2025), taxonomies for digital building passports Vangelova et al. (2026). What none of these applications do is

open the record and look inside. That is the step taken here: topic inference when a field is missing, field-restricted read-out, and a check of whether such read-outs can be measured reliably.

We make three contributions.

1. **A field-cooperative neural topic model with native missing-field support.** FC-NTM models the two fields of a record as two conditional variational streams that share one linear topic decoder; cross-field reconstruction and a field consistency regulariser encode the field relation into a shared topic space, and field-level word masking with whole-field dropout lets one trained model serve complete, partially observed and title-only records without retraining or model surgery.
2. **A unified topic–word read-out protocol for comparative evaluation.** All models are scored on one evaluation vocabulary with exactly ten words per topic, each model being read out both through its native topic–word matrix and through a class term-frequency–inverse-document-frequency (TF–IDF) read-out. This separates differences caused by the read-out from differences caused by the model — a common confound in topic-model benchmarking — and is what makes the reported comparisons interpretable.
3. **Multi-seed evidence on when field cooperation helps, and an engineering screening task.** Ten-seed paired comparisons give reproducible read-out gains on the civil engineering corpus (+0.049 class TF–IDF and +0.061 native read-out, 10/10 seeds) and on GitHub issues (+0.028, 9/10), but not on AG News (−0.009, 3/10), while a same-budget concatenated-field control locates the gains in the complete configuration rather than in the two-stream architecture alone. A title-only routing experiment on the civil corpus (0.66 accuracy on the evaluated subset) shows how the trained model can screen engineering records from titles alone.

Figure 1 summarises the problem setting and the three contributions: dual-field scholarly records whose abstracts may be missing, the field-cooperative configuration that trains both fields against one shared decoder, and the observation regimes a single trained model has to serve.

2 Related work

Neural topic models.

Neural topic models map a bag of words to a latent Gaussian posterior and decode topics from it. NVDM was the early application of variational auto-encoders to text Miao et al. (2016); AVITM brought autoencoding variational inference and, through its ProdLDA variant, a linear decoder whose topic–word matrix can be read directly Srivastava and Sutton (2017). CTM and CombinedTM concatenate sentence embeddings with bag-of-words counts Bianchi et al. (2021), ETM places topics in an embedding space Dieng et al. (2020), and for short texts BTM keeps to word pairs Yan et al. (2013). BERTopic builds its topic representations from class TF–IDF Grootendorst (2022); Top2Vec learns joint document, word and topic embeddings Angelov (2020). GraphBTM and BCK-TM add word-pair structure Zhu et al. (2018); Zhang et al.

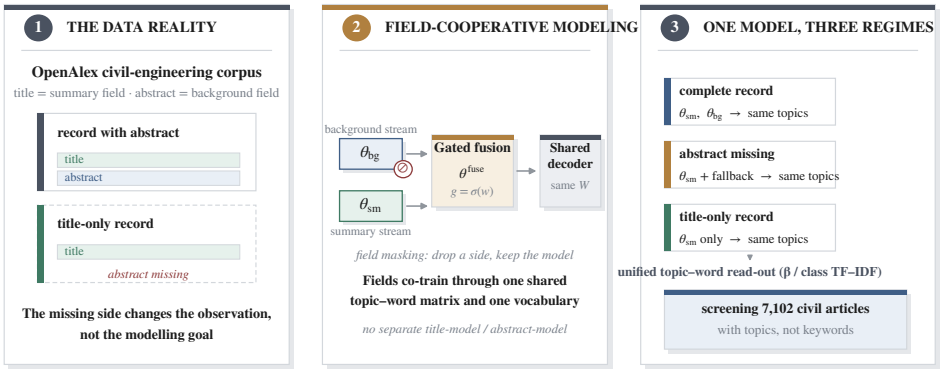


Fig. 1 Problem setting and contributions. (1) The application corpus is dual-field scholarly metadata in which the abstract (background field) may be missing, leaving title-only records. (2) FC-NTM encodes both fields with separate conditional variational streams tied to one shared topic–word decoder; field masking drops a side at training or inference time without changing the model. (3) One trained model serves complete, abstract-missing and title-only records through a unified topic–word read-out, supporting topic-based screening of civil-engineering literature.

(2022), and BigARTM contributes an additive regularisation framework Vorontsov et al. (2015). Topic-distribution quantisation with a negative-sampling decoder is a more recent short-text option Wu et al. (2020). Our own decoder follows ProdLDA, with topic orthogonality, dropout Srivastava et al. (2014) and validation-based early stopping Prechelt (1998); beyond that, objectives, training loops and learning rates differ across the algorithms.

Multi-view, multimodal and structured-document models.

Sharing information across views is the business of multi-view latent-variable learning: multi-source variational inference Kurle et al. (2019), robust multi-view topic modelling Zhang et al. (2017), and the disentangled or dual-view topic models Pergola et al. (2021); Liu et al. (2022). Structured neural topic models for reviews Esmaeili et al. (2019) and metadata-conditioned models Card et al. (2018); Roberts et al. (2019) exploit other kinds of document structure. CRNTM and DVI-DTM are recent short-text examples Feng et al. (2022); Liu et al. (2026), with related word-pair models nearby Li et al. (2019); Zhu et al. (2021). Kurle et al. encode each source separately and align the posterior representations, JMVAE-kl infers a missing modality from those that remain Suzuki et al. (2016), and SubNTM models sub-document topics Nguyen et al. (2026). What fixes our setting is the record schema: two textual fields, one shared vocabulary, one linear decoder — and a question none of the above asks, namely what field-restricted read-outs and missing-background inference actually measure. The established components are simply put to work at the field level.

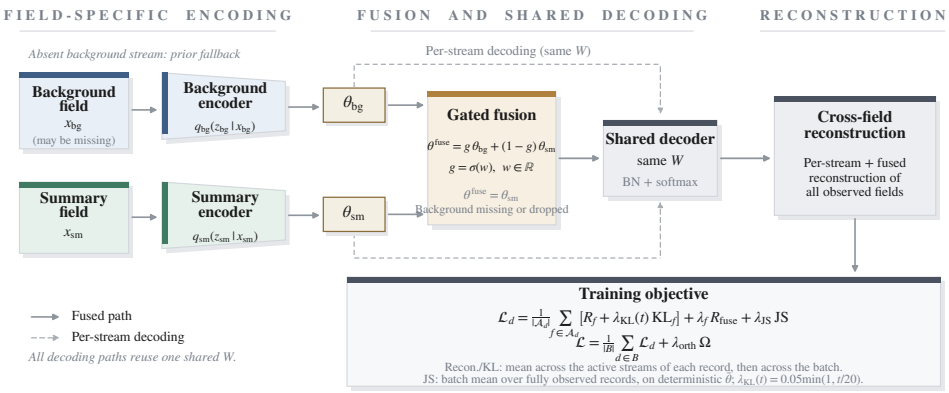


Fig. 2 Overview of FC-NTM. Each field has its own conditional variational stream. Solid arrows show the fused path, in which the gated mixture is decoded and every present field is reconstructed; dashed arrows show the per-stream decodes, which reuse the same decoder parameters. The training objective combines per-stream reconstruction, the KL terms of the active streams, fused reconstruction, the JS term (computed only when both streams are active) and the topic-orthogonality penalty. A missing or dropped field falls back to the prior, so the gated mixture reduces to the remaining stream ($\theta^{\text{fuse}} = \theta_{\text{sm}}$ when the background field is missing or dropped). The native β read-out uses the raw matrix W, whereas generation follows the batch-normalised path of Eq. (2).

Text mining and topic modelling of engineering documents.

Within engineering informatics the text-mining literature is extensive — safety reports Tixier et al. (2016); Zhang et al. (2019), regulatory and contractual documents Salama and El-Gohary (2016); Zhou and El-Gohary (2016); Moon et al. (2022); Pham and Han (2023), domain surveys Wu et al. (2022); Shamshiri et al. (2024) — and topic models appear in it as DUET Wang and Taylor (2019), as a study of machine-learning applications in buildings Zhou et al. (2025) and as passport taxonomies Vangelova et al. (2026). All of it treats a record as one flat text. The question taken up here is different: what the record's fixed field structure does to field-restricted read-out and to missing-field inference.

3 Methodology

Figure 2 gives an overview of FC-NTM. The two fields are encoded by separate conditional variational streams that share one topic decoder; the streams are tied by cross-field reconstruction and a consistency term, and field-level masking with whole-field dropout covers partially and fully missing fields.

3.1 Problem setting

Each record d consists of field A (*background*, e.g. the abstract of a paper or the body of an issue) and field B (*summary*, e.g. the title), with N records in total and a shared evaluation vocabulary of V terms. The model maintains K topics, a document–topic distribution $\theta_d \in \Delta^{K-1}$, and a topic–word matrix $\beta \in \mathbb{R}^{K \times V}$ shared by both fields. The task is to infer θ_d and β from unlabelled records, to produce readable topics in

two read-outs, and to remain usable when one field is partially or entirely missing at inference time.

3.2 Field streams and the shared decoder

Each field is encoded by its own two-layer MLP encoder (200 hidden units, Softplus activations, dropout 0.2) into the parameters of a field-conditioned Gaussian posterior,

$$q_f(z_f \mid x_f) = \mathcal{N}\big(\mu_f(x_f), \mathrm{diag}(\sigma_f^2(x_f))\big), \qquad f \in \{\mathrm{bg}, \mathrm{sm}\}, \tag{1}$$

where x_f is the L_1-normalised bag of words of field f and $\log \sigma_f^2$ is capped at 2. Posterior means are mapped to topic proportions by a deterministic softmax, $\bar{\theta}_f = \mathrm{softmax}(\mu_f(x_f))$, which is used at evaluation time in place of sampling; $\bar{\theta}_f$ is a deterministic topic-proportion estimate and does not equal the expectation of $\mathrm{softmax}(z_f)$. Both streams share one linear topic–word matrix $W \in \mathbb{R}^{K \times V}$, and we write $\theta_f \in \mathbb{R}^{1 \times K}$ for the topic proportions of field f under the row-vector convention,

$$p(w \mid \theta_f) = \mathrm{softmax}\big(\mathrm{BN}(\theta_f W)\big), \tag{2}$$

where θ_f passes through dropout before the product, BN is a batch-normalisation layer *without* affine parameters applied over the vocabulary dimension (batch statistics during training, running statistics at evaluation), and there is no additive bias. The β read-out of Section 3.6 reads the raw rows of W, whereas generation uses the normalised path of Eq. (2); the gap between the two is quantified in the Supplementary Material.

3.3 Training objective

For a document d let $\mathcal{G}_d$ collect its present fields (the background field is present when the record carries it; the summary stream is always instantiated, including zero-vector inputs after preprocessing) and let $\mathcal{A}_d \subseteq \mathcal{G}_d$ be the *active streams*, i.e. present fields whose stream is not removed by whole-field dropout. Write $n_{dw}^{(g)}$ for the raw count of word w in field g. Each active stream decodes once (Eq. (2)) and reconstructs *every present field*,

$$R_f = \sum_{g \in \mathcal{G}_d} \omega_g \, R_f(g), \qquad R_f(g) = -\sum_{w=1}^{V} n_{dw}^{(g)} \log p(w \mid \theta_f)\,, \tag{3}$$

with fixed field weights $\omega_{\mathrm{bg}} = 2.0$ and $\omega_{\mathrm{sm}} = 1.0$. The summary stream reconstructs the background field as well (cross-field reconstruction), and a background field that was observed but dropped during training stays in the reconstruction targets even though its own stream is inactive, whereas a naturally missing background field enters neither the inputs nor the targets. The per-document objective is

$$\mathcal{L}_d = \frac{1}{|\mathcal{A}_d|} \sum_{f \in \mathcal{A}_d} \Big[R_f + \lambda_{\mathrm{KL}}(t)\, \mathrm{KL}_f \Big] + \lambda_f R_{\mathrm{fuse}} + \lambda_{\mathrm{JS}}\, \mathrm{JS}, \tag{4}$$

where $\mathrm{KL}_f = \mathrm{KL}\big(q_f(z_f \mid x_f) \,\|\, \mathcal{N}(0, I)\big)$ is summed over the K latent dimensions and clamped at zero, and $\lambda_{\mathrm{KL}}(t) = 0.05 \min(1, t/20)$ anneals linearly during the first 20 epochs. The fused term $R_{\mathrm{fuse}} = \sum_{g \in \mathcal{G}_d} \omega_g R_{\mathrm{fuse}}(g)$ is a second reconstruction of both present fields from the fused topic mixture

$$\theta_d^{\mathrm{fuse}} = g\,\theta_{\mathrm{bg}} + (1-g)\,\theta_{\mathrm{sm}}, \qquad g = \sigma(w), \quad w \in \mathbb{R}. \tag{5}$$

for records with both fields present, and $\theta_d^{\mathrm{fuse}} = \theta_{\mathrm{sm}}$ when the background field is missing or dropped. During training Eq. (5) combines the *sampled* per-stream proportions that also feed the per-stream reconstructions; at inference time, and for the JS term below, the deterministic estimates $\bar{\theta}_f$ are used. At inference, Eq. (5) is a convex combination of two deterministic topic-proportion estimates. We refer to this quantity as a *fused topic mixture* to distinguish it from a posterior. The gate g receives gradients only through R_{fuse}, so with $\lambda_f = 0$ the model degenerates exactly to fixed mean fusion. The consistency term

$$\mathrm{JS} = \tfrac{1}{2}\,\mathrm{KL}\big(\bar{\theta}_{\mathrm{bg}} \| m\big) + \tfrac{1}{2}\,\mathrm{KL}\big(\bar{\theta}_{\mathrm{sm}} \| m\big), \qquad m = \tfrac{1}{2}\big(\bar{\theta}_{\mathrm{bg}} + \bar{\theta}_{\mathrm{sm}}\big), \tag{6}$$

acts on the deterministic topic proportions and does not use the Gaussian posteriors. It is computed only for documents whose two streams are both active and is not applied to records with one active stream. A topic-orthogonality penalty keeps the rows of W apart,

$$\Omega = \frac{1}{K(K-1)} \sum_{k \neq k'} \big|\cos\langle W_k, W_{k'}\rangle\big|, \tag{7}$$

and the batch objective is $\frac{1}{|B|}\sum_{d \in B} \mathcal{L}_d + \lambda_{\mathrm{orth}}\Omega$ with $\lambda_{\mathrm{orth}} = 0.2$.

3.4 Field masking

Two masking mechanisms expose the model to missing fields, and they act on different units. (i) *Word-level input dropout* (denoising) is applied during training to the bag-of-words *word types* of each field: each bag-of-words dimension is independently zeroed with probability 0.30 (background) or 0.40 (summary) in the encoder view only, while the reconstruction targets remain the complete counts. (ii) *Whole-field dropout* removes the background stream of each training document with probability 0.5 and substitutes the prior; the summary stream is never dropped and learns to reconstruct both fields on its own. The test-time occlusion protocol of Section 5.5 masks *token positions* in the raw field text. The training and evaluation procedures use different masking units. The consistency weight $\lambda_{\mathrm{JS}} = 0.5$, the fusion weight $\lambda_f = 0.5$ and the dropout probabilities were selected on held-out data.

3.5 Optimisation

Training uses Adam (learning rate 5×10^{-4}, momentum parameters 0.9 and 0.999), mini-batches of 128, gradient-norm clipping at 5.0 and a budget of 700 epochs. After the KL warm-up the state with the lowest held-out loss is kept, for which 20% of the training split is held out; the patience is 150 epochs. Evaluation-mode inference is

deterministic: the reparameterisation sampling is switched off and the held-out loss uses posterior means. Numerical-anomaly recovery rolls the weights back to the most recent finite epoch, clears the optimiser moments and halves the learning rate; this recovery step is separate from the best-held-out-epoch selection of the early-stopping loop.

3.6 Unified topic–word read-out

Topic-word comparisons are only meaningful if every model is read out with the same vocabulary and the same number of words per topic. We fix one evaluation vocabulary and exactly ten words per topic. The β **read-out** takes the top ten words of each normalised row of the model's topic–word matrix; for FC-NTM these are the raw rows of W, whereas generation uses Eq. (2). LDA, BTM, NMF, ProdLDA, ETM, CTM-lite, CombinedTM and FC-NTM are all evaluated this way. The **class TF–IDF read-out** aggregates the training-term counts by the soft document–topic assignment of the model, applies a BERTopic-style class TF–IDF transform Grootendorst (2022), and then takes the top ten words. Soft assignments replace the hard cluster membership used in the standard class-level aggregation; for FC-NTM this read-out is built from the fused topic mixture $\bar{\theta}^{\text{fuse}}$, which is why the two read-outs can rank models differently. The sensitivity of the raw-parameter read-out relative to the generative read-out is reported in the Supplementary Material.

4 Experimental setup

4.1 Corpora and evaluation protocol

We use three dual-field corpora. The **civil engineering corpus** is the application corpus: title–abstract records of English-language scholarly works from OpenAlex Priem et al. (2022) (https://openalex.org) in *Civil and Structural Engineering* and *Building and Construction* (2000–2023; CC BY, CC BY-SA, or public-domain records). The sample comprises 7,102 articles, 889 conference papers and 9 reviews, split into 6000 training and 2000 test records, with the title as the summary field and the abstract as the background field (mean length ratio 11.2). **GitHub issues** OpenIndex (2026) pairs the issue body with its title (four mature repositories, 6000/2000, ratio 12.9), and **AG News** Zhang et al. (2015) pairs the description with the headline (6000/7600, ratio 3.9). All corpora are lower-cased and tokenised on letter sequences; stop words and very short tokens are removed, the vocabulary is built from the training split only, and test records never enter the vocabulary. Table 3 reports the application corpus and Tables 1 and 2 report the two public corpora, all on the same evaluation vocabulary with exactly ten words per topic and $K = 30$. The evaluation protocol (vocabulary, ten words per topic, coherence and diversity metrics) is shared by every model, whereas the training procedure differs across algorithm families and is described in Section 4.2. Because 400 records from each public test corpus were used during design, every public-corpus result is also recomputed on a development-free subset (1600 GitHub and 7200 AG News records). This rescoring excludes direct test overlap and is reported as a sensitivity analysis; Section 6 considers the remaining effect of prior

Table 1 Topic quality on AG News (unified evaluation vocabulary, ten words per topic; values are means over seeds 11/23/37 where applicable, and the training procedure differs across algorithm families as described in Section 4.2; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.600±0.011	0.164±0.005	−4.19	0.409	0.980
CombinedTM	30	0.526±0.037	0.057±0.070	−8.01	0.118	1.000
FC-NTM-cTFIDF	30	**0.612**±0.014	**0.173**±0.012	−4.20	0.193	0.974
FC-NTM	30	0.586±0.026	0.137±0.022	−4.91	0.013	0.993
ETM	30	0.506±0.014	0.092±0.010	−5.12	0.439	0.982
ProdLDA	30	0.526±0.031	0.071±0.015	−6.61	0.055	0.999
Top2Vec	2.7	0.498±0.032	0.099±0.031	−2.92	0.268	0.937
NMF	30	0.494±0.015	0.069±0.013	−5.96	0.799	0.997
BERTopic	29	0.488±0.022	0.035±0.026	−8.81	0.681	0.997
CTM-lite	30	0.431±0.023	−0.015±0.016	−8.34	0.041	0.998
LDA	30	0.421±0.013	−0.066±0.018	−8.62	0.696	0.994

Table 2 Topic quality on GitHub issues (same evaluation protocol and read-outs as Table 1; values are means over seeds 11/23/37; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.712±0.012	0.232±0.012	−1.79	0.349	0.934
FC-NTM-cTFIDF	30	**0.727**±0.022	**0.274**±0.026	−1.90	0.133	0.901
FC-NTM	30	0.701±0.016	0.233±0.012	−3.16	0.010	0.967
ETM	30	0.675±0.014	0.213±0.010	−2.14	0.366	0.955
CombinedTM	30	0.643±0.022	0.193±0.015	−4.70	0.031	0.983
NMF	30	0.614±0.005	0.164±0.003	−3.26	0.595	0.985
ProdLDA	30	0.579±0.007	0.107±0.012	−5.21	0.043	0.995
CTM-lite	30	0.585±0.032	0.129±0.036	−5.11	0.011	0.989
LDA	30	0.546±0.011	0.074±0.017	−4.21	0.684	0.990
BERTopic	29	0.540±0.015	0.076±0.012	−7.00	0.546	0.976
Top2Vec	160	0.493±0.009	−0.008±0.013	−9.65	0.337	0.994

model selection. The results are organised around three questions: how well civil title–abstract records can be structured into verifiable topics; how much signal remains when the abstract is missing; and how much the complete field-cooperative configuration adds over a concatenated single-encoder control. Cross-corpus diagnostics round off the analysis.

Table 3 Topic quality on the civil engineering corpus (unified evaluation vocabulary, ten words per topic; values are means over seeds 11/23/37, and the training procedure differs across algorithm families as described in Section 4.2; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.583±0.010	0.128±0.009	−2.18	0.283	0.944
CombinedTM	30	0.539±0.029	0.082±0.028	−6.11	0.063	0.998
FC-NTM-cTFIDF	30	**0.636**±0.014	**0.159**±0.011	−2.21	0.165	0.937
FC-NTM	30	0.633±0.014	0.154±0.022	−2.85	0.014	0.990
ETM	30	0.581±0.012	0.119±0.007	−2.75	0.379	0.963
ProdLDA	30	0.580±0.023	0.125±0.017	−3.84	0.051	0.999
Top2Vec	153	0.515±0.002	0.029±0.011	−8.52	0.359	0.995
NMF	30	0.598±0.009	0.135±0.006	−2.86	0.559	0.984
BERTopic	29	0.576±0.026	0.116±0.027	−5.06	0.563	0.991
CTM-lite	30	0.536±0.030	0.098±0.027	−3.85	0.013	0.995
LDA	30	0.554±0.008	0.056±0.025	−4.62	0.602	0.987

4.2 Baselines, training protocols and metrics

Baselines are LDA Blei et al. (2003), BTM Yan et al. (2013), NMF with a KL objective Lee and Seung (2001), ProdLDA Srivastava and Sutton (2017), ETM Dieng et al. (2020), CTM-lite, the official CombinedTM Bianchi et al. (2021), BERTopic Grootendorst (2022) and Top2Vec Angelov (2020); contextual models use Sentence-BERT embeddings Reimers and Gurevych (2019). The implementations fall into four families with different training procedures: Gibbs sampling (BTM, LDA), matrix factorisation (NMF), neural variational models (ProdLDA, ETM, CTM-lite and FC-NTM share the Adam loop Kingma and Ba (2015) and the 700-epoch budget but differ in learning rate; the official CombinedTM keeps its own loop), and embedding clustering (BERTopic, Top2Vec), which choose their own topic numbers and are shown for reference only. Full hyper-parameters are listed in the Supplementary Material. Quality is measured by C_V coherence Röder et al. (2015); NPMI Bouma (2009); Lau et al. (2014); UCI and UMass Newman et al. (2010); Mimno et al. (2011); inter-topic diversity as one minus the mean rank-biased overlap (RBO) with persistence 0.9 and the residual tail retained for fixed ten-word lists Webber et al. (2010); and an exclusivity score based on Bischof and Airoldi Bischof and Airoldi (2012), normalised by K so that zero corresponds to the chance-level value $1/K$. Topics with non-finite scores are dropped, so the effective topic count is recorded next to every score.

5 Results and discussion

5.1 Main results

On the civil engineering corpus (Table 3), FC-NTM attains the highest coherence among all evaluated models under both read-outs ($C_V = 0.633$ for the β read-out and

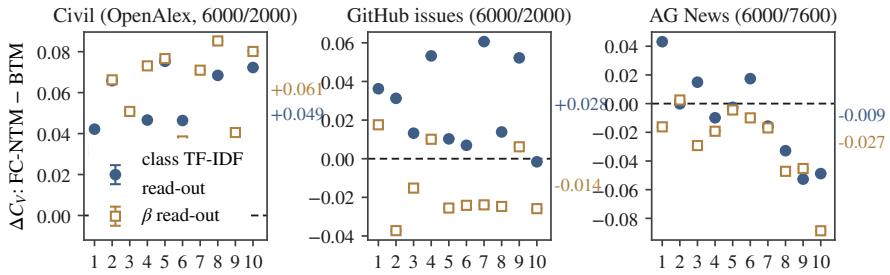


Fig. 3 Per-seed paired comparison of FC-NTM and BTM (ten seeds per corpus, C_V). Each point is one paired seed; the dashed line marks the BTM level. The class TF–IDF read-out is positive in all ten paired seeds on the civil corpus and in nine of ten on GitHub issues, whereas the AG News seeds are mixed. This is the corpus-dependent pattern that the mean differences summarise.

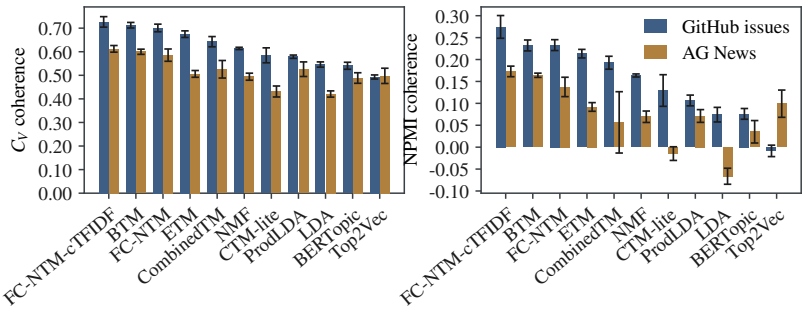


Fig. 4 Main results on the two public corpora (unified evaluation vocabulary, ten words per topic). Left: C_V; right: NPMI. Error bars are standard deviations across seeds 11/23/37.

0.636 for the class TF–IDF read-out; NPMI 0.154 and 0.159 over seeds 11/23/37), ahead of NMF (0.598), BTM (0.583) and the neural baselines, and the ten-seed extension keeps both read-out advantages (Table 5); Figure 3 shows the per-seed differences behind these means across all three corpora. Figure 4 compares all evaluated models on the two public corpora under both coherence metrics. On AG News, FC-NTM-cTFIDF reaches $C_V = 0.612$ and NPMI = 0.173, above BTM under the matched class TF–IDF read-out (0.593 / 0.169) and above the β read-outs of ProdLDA, ETM, NMF, CTM-lite and LDA; under its own β read-out BTM keeps a small lead (0.600 versus 0.586). On GitHub issues, FC-NTM-cTFIDF records $C_V = 0.727$ and NPMI = 0.274, above BTM at both read-outs (0.712 / 0.232 for the native β read-out, 0.700 / 0.235 matched), and above the sentence-embedding baselines (CombinedTM 0.643, BERTopic 0.540). The matched-read-out advantage over BTM is +0.027 C_V and +0.039 NPMI on GitHub and +0.019 / +0.004 on AG News for three seeds, but the per-seed structure differs: on GitHub all three seeds are positive, whereas the AG News mean is carried by two of three seeds and the NPMI advantage by one. On the development-free test subset the ordering is unchanged (GitHub 0.734 versus 0.711, NPMI 0.284 versus 0.248; AG News 0.610 versus 0.590, NPMI 0.172 versus 0.167; with BTM's native β read-out, 0.734 versus 0.726 and 0.610 versus 0.598).

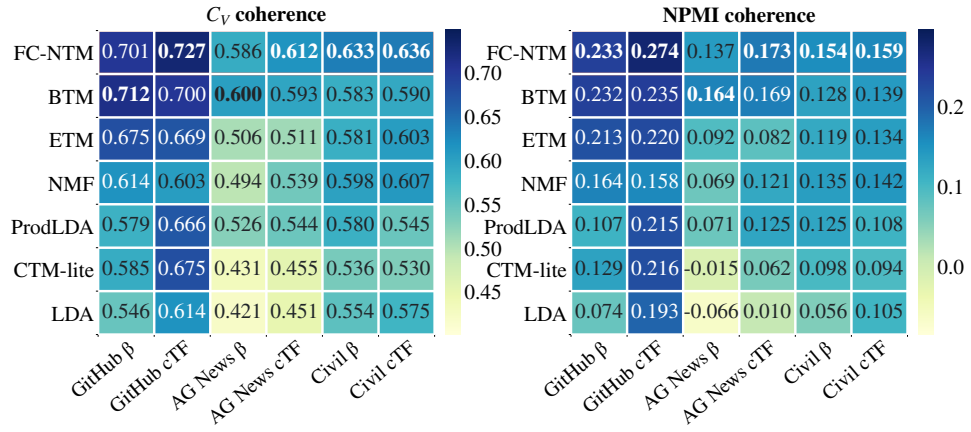


Fig. 5 The read-out comparison of Table 4 as a heatmap ($K = 30$, means over seeds 11/23/37). Left: C_V; right: NPMI. Columns combine the three corpora with the two read-outs (native β; class TF–IDF). The boxed first row is FC-NTM; bold entries mark the column-wise best.

5.2 Read-out-matched comparison

Table 4 collects the read-out-matched comparison across models and read-outs.

The class TF–IDF read-out depends on the model and the corpus: on GitHub it lifts CTM-lite, ProdLDA and LDA by 0.091, 0.087 and 0.068 but leaves ETM and NMF unchanged or worse, while on AG News it helps every model except BTM. Comparisons use the matched read-out. Under it, FC-NTM-cTFIDF leads the strongest GitHub baselines (BTM 0.700, CTM-lite 0.675, ETM 0.669, ProdLDA 0.666) by 0.027–0.061 C_V and 0.039–0.059 NPMI, larger than its seed-to-seed standard deviation (0.022); on AG News the advantage over BTM is +0.019 C_V but rests on two of three seeds; under the native β read-out the AG News means are 0.586 for FC-NTM and 0.600 for BTM. On the civil corpus NMF (0.607) and ETM (0.603) exceed BTM (0.590), and FC-NTM-cTFIDF leads them by 0.029 and 0.033 C_V, although the ten-seed comparison is against BTM. Figure 5 visualises the same comparison; within each panel one colour scale spans all corpora and read-outs, the boxed first row is FC-NTM, and bold entries mark the best model per column.

5.3 Ten-seed paired comparison

To test whether the three-seed differences are seed noise, we extend FC-NTM and BTM to ten seeds on all three corpora (adding 53/71/89/101/113/127/139) and compare both read-outs and the occlusion behaviour seed by seed. Occlusion comparisons are limited to preselected baselines: BTM on all three corpora and LDA additionally on AG News.

For GitHub, the class TF–IDF difference is +0.028 C_V (95% CI $[+0.012, +0.043]$, 9/10 positive seeds), and the relative-degradation difference is +0.024 (10/10). On the civil corpus, the class TF–IDF and native β differences are +0.049 and +0.061 C_V, with 10/10 positive paired differences; the occlusion difference is −0.003 (0/10). On AG News, the class TF–IDF difference is −0.009 (95% CI $[-0.030, +0.013]$, 3/10) and

Table 4 Read-out comparison ($K = 30$, means over seeds 11/23/37). Each model is first read out through its own topic–word matrix (β read-out) and then through a class TF–IDF read-out aggregated from its training-set posterior θ; BTM enters the same read-out through the point-estimate document–topic posterior of its original Section 3.3, and the FC-NTM row is the same model as in Tables 1 and 2.

Corpus	Model	β read-out C_V / NPMI	Class TF–IDF C_V / NPMI
GitHub	FC-NTM(3 seed mean)	0.701 / 0.233	**0.727 / 0.274**
	ETM	0.675 / 0.213	0.669 / 0.220
	ProdLDA	0.579 / 0.107	0.666 / 0.215
	CTM-lite	0.585 / 0.129	0.675 / 0.216
	NMF	0.614 / 0.164	0.603 / 0.158
	LDA	0.546 / 0.074	0.614 / 0.193
	BTM(3 seeds)	0.712 / 0.232	0.700 / 0.235
AG News	FC-NTM(3 seed mean)	0.586 / 0.137	**0.612 / 0.173**
	ProdLDA	0.526 / 0.071	0.544 / 0.125
	NMF	0.494 / 0.069	0.539 / 0.121
	ETM	0.506 / 0.092	0.511 / 0.082
	CTM-lite	0.431 / −0.015	0.455 / 0.062
	LDA	0.421 / −0.066	0.451 / 0.010
	BTM(3 seeds)	0.600 / 0.164	0.593 / 0.169
Civil corpus	FC-NTM(3 seed mean)	**0.633 / 0.154**	**0.636 / 0.159**
	NMF	0.598 / 0.135	0.607 / 0.142
	ETM	0.581 / 0.119	0.603 / 0.134
	BTM(3 seeds)	0.583 / 0.128	0.590 / 0.139
	LDA	0.554 / 0.056	0.575 / 0.105
	ProdLDA	0.580 / 0.125	0.545 / 0.108
	CTM-lite	0.536 / 0.098	0.530 / 0.094

Table 5 Ten-seed paired comparison of FC-NTM and BTM on the three corpora. The read-out rows report C_V differences (FC-NTM minus BTM at the same read-out); the occlusion rows compare the relative degradation NLL(100%)/NLL(25%), so a positive value means FC-NTM is more stable. Occlusion comparisons are limited to preselected baselines: BTM on all three corpora and LDA additionally on AG News. CIs are two-sided 95% t intervals; sign and Wilcoxon tests are two-sided.

Corpus	Comparison	Mean diff.	CI low	CI high	Positive	Sign p	Wilcoxon p
GitHub	Class TF–IDF	+0.028	+0.012	+0.043	9/10	0.021	0.004
GitHub	β read-out	−0.014	−0.028	−0.001	3/10	0.344	0.037
GitHub	Occlusion (vs. BTM)	+0.024	+0.022	+0.026	10/10	0.002	0.002
AG News	Class TF–IDF	−0.009	−0.030	+0.013	3/10	0.344	0.375
AG News	β read-out	−0.027	−0.047	−0.008	1/10	0.021	0.004
AG News	Occlusion (vs. BTM)	+0.025	+0.024	+0.026	10/10	0.002	0.002
AG News	Occlusion (vs. LDA)	−0.013	−0.014	−0.012	0/10	0.002	0.002
Civil corpus	Class TF–IDF	+0.049	+0.033	+0.064	10/10	0.002	0.002
Civil corpus	β read-out	+0.061	+0.047	+0.075	10/10	0.002	0.002
Civil corpus	Occlusion (vs. BTM)	−0.003	−0.004	−0.002	0/10	0.002	0.002

the native β difference is −0.027 (1/10). The native β difference on GitHub is −0.014 (3/10). The class TF–IDF mean differences across the three corpora are +0.049, +0.028 and −0.009. Confidence intervals describe training and masking randomness under a fixed corpus and design.

Table 6 Field redundancy, measured occlusion gap and the C_V difference against BTM on two corpora. Redundancy is fitted on the training split and evaluated on the test split and includes constructive controls (exact copy = 1.000, cross-document shuffle ≈ 0.000); the occlusion ΔC_V is the mean±standard deviation over seeds 11/23/37 with dropout training. The last column gives the C_V difference under the matched class TF–IDF read-out and under BTM's native β read-out.

Corpus	Held-out Ridge R^2	Held-out ρ^2_{cca}	Occlusion ΔC_V	C_V gap, cTFIDF−BTM (matched / β)
GitHub issues	0.230	0.306	+0.009±0.022	+0.027 / +0.014
AG News	0.158	0.245	+0.018±0.013	+0.019 / +0.012

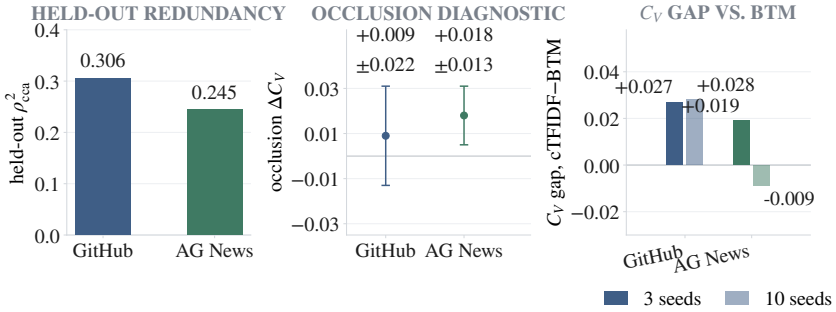


Fig. 6 Field redundancy and gain (held-out measures). Left: squared canonical correlation ρ^2_{cca} (higher means more redundant), fitted on the training split and evaluated on the test split. Middle: occlusion diagnostic ΔC_V (mean±standard deviation over seeds 11/23/37). Right: the C_V difference of FC-NTM-cTFIDF against BTM under the matched class TF–IDF read-out (three seeds) and the ten-seed paired difference. The two corpora reach similar held-out redundancy, while the ten-seed difference is +0.028 on GitHub and −0.009 on AG News.

5.4 Field redundancy and the gain

Table 6 reports the held-out redundancy measures together with the observed gains, and Figure 6 relates them across the two public corpora.

The redundancy measures are fitted on the training split and evaluated on the test split for the two public corpora. Their held-out predictability is 0.306 for GitHub and 0.245 for AG News (ρ^2_{cca}). At three seeds, the matched class TF–IDF differences from BTM are +0.027 and +0.019; the ten-seed differences are +0.028 (9/10) and −0.009 (3/10). The corresponding civil difference is +0.049 (10/10), as reported in Table 5. Whole-field occlusion changes coherence by $+0.009 \pm 0.022$ on GitHub and $+0.018 \pm 0.013$ on AG News.

5.5 Field inference under partial occlusion

A direct test of field complementarity is whether the visible field B helps a model recover content when part of field A is unavailable. The protocol (Supplementary Material) masks a fraction p of the background-field tokens in the test documents, re-infers the topic proportions from the remaining input, and scores the occluded tokens under the model's own word distribution; it reports the per-token negative

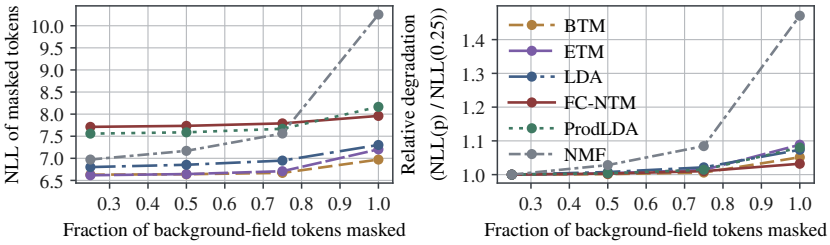


Fig. 7 Inference behaviour under partial occlusion of the background field (GitHub issues, seed 11, 2000 test documents). Left: per-token negative log-likelihood of the occluded tokens; right: relative degradation NLL(p)/NLL(0.25), which summarises the relative increase from the 25% condition. FC-NTM is trained with whole-field dropout at $p = 0.5$ and has the smallest relative degradation ($\times$1.032). The figure also reports absolute NLL to contextualise this ratio; topic agreement is reported in Table 7.

Table 7 Partial-occlusion curves on GitHub issues (seed 11). NLL is the per-token negative log-likelihood of the occluded tokens (lower is better); relative degradation is NLL(100%)/NLL(25%); agreement is the fraction of documents holding their most probable topic at 100% occlusion. BTM enters through its document–topic posterior.

Model	NLL(25%)	NLL(50%)	NLL(75%)	NLL(100%)	Degradation / Agreement
FC-NTM	7.712	7.735	7.790	7.960	**1.032** / 0.358
LDA	6.802	6.854	6.948	7.302	1.074 / 0.420
ProdLDA	7.558	7.588	7.668	8.165	1.080 / 0.354
BTM	6.630	6.639	6.670	**6.972**	1.052 / 0.448
ETM	**6.616**	6.643	6.708	7.202	1.089 / 0.452
NMF	6.972	7.169	7.561	10.255	1.471 / **0.469**

log-likelihood (NLL), the agreement of the most probable topic with the unoccluded inference, and the posterior JS divergence. Every model uses its own inference path and its own word distribution, so no cross-model read-out convention is involved. When a document has no visible evidence at all, BTM and NMF use the training-corpus topic prior; treating a zero vector as a word distribution would inflate the NLL to $\log 10^{12} = 27.63$. Table 7 reports the per-condition values on GitHub issues, and Figure 7 plots the NLL and relative-degradation curves behind them.

From 25% to 100% occlusion the per-token NLL of FC-NTM rises by only 3.2% (7.712 $\rightarrow$ 7.960), less than LDA (7.4%), ProdLDA (8.0%), ETM (8.9%), BTM (5.2%) and NMF (47.1%), which is consistent with the training-time whole-field dropout. At full occlusion, BTM has the lowest NLL (6.972); topic agreement is highest for NMF (0.469), followed by ETM (0.452), while FC-NTM reaches 0.358. Relative degradation is therefore always read alongside absolute NLL and agreement, because the ratio alone says nothing about how good the absolute predictions are. The protocol covers three English corpora, ten seeds per corpus, occlusion masks redrawn for every seed, and the field split follows the existing title/body structure.

Table 8 Partial-occlusion behaviour on the three corpora (ten seeds 11–139, mean±standard deviation; the mask is redrawn per seed). Bold marks the smallest relative degradation per corpus. Field asymmetry is described by the mean background/summary token ratio and by the token-type share of background words that do not occur in the summary of the same document.

Corpus(Field asymmetry)	Model	Relative degradation	Agreement(100%)
GitHub issues Length ratio 12.9 Background-exclusive 0.877	FC-NTM	**1.028±0.003**	0.364
	LDA	1.075±0.003	0.416
	ProdLDA	1.078±0.004	0.343
	ETM	1.087±0.003	0.451
	BTM	1.052±0.001	0.443
	NMF	1.467±0.009	0.464
Civil corpus Length ratio 11.2 Background-exclusive 0.894	FC-NTM	1.010±0.001	0.661
	LDA	1.037±0.002	0.574
	ProdLDA	1.028±0.002	0.558
	ETM	1.041±0.001	0.656
	BTM	**1.007±0.001**	0.565
	NMF	1.172±0.002	0.651
AG News Length ratio 3.9 Background-exclusive 0.868	FC-NTM	1.019±0.001	0.515
	LDA	**1.006±0.002**	0.440
	ProdLDA	1.026±0.001	0.491
	ETM	1.072±0.002	0.561
	BTM	1.044±0.001	0.553
	NMF	1.411±0.004	0.545

5.6 Cross-corpus comparison

Table 8 and Figure 8 extend the same protocol to the three corpora.

On GitHub (ratio 12.9), all models degrade by factors of ×1.028–×1.467, and FC-NTM has the smallest relative degradation; the ten-seed paired difference from BTM is +0.024±0.004 (10/10). On the civil corpus (ratio 11.2), the relative degradations are 1.010 for FC-NTM and 1.007 for BTM (paired difference −0.003, 0/10). On AG News (ratio 3.9), they are 1.019 for FC-NTM and 1.006 for LDA. Held-out mutual prediction is slightly higher for GitHub, while the length ratio and background-exclusive share distinguish GitHub from AG News and place the civil corpus in a similar range to GitHub. These statistics serve as corpus descriptors for the three evaluated collections.

5.7 Field-restricted read-out

Field-restricted read-out truncates the topic words to one field's vocabulary and scores them against that field's own test text. With the shared decoder, the background-view C_V of FC-NTM stays close to its full-vocabulary β read-out on all three corpora (0.697, 0.566 and 0.637 versus 0.701, 0.586 and 0.633), while the summary view is lower in absolute terms mainly because a summary contains only 3–5 tokens and the reference corpus is very short, so that column should be compared within the same protocol. The cross-model comparison shows that the read-out is not unique to a two-stream encoder: BTM is slightly better in the background view on GitHub and AG News (0.710 versus 0.697 and 0.586 versus 0.566), FC-NTM leads on the civil corpus (0.637 versus 0.600 for NMF), and in the summary view LDA and ProdLDA are above

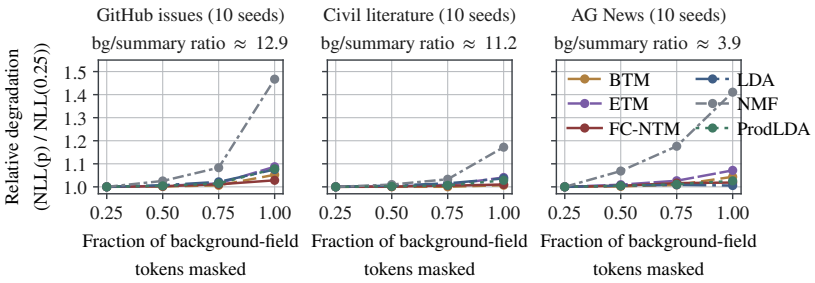


Fig. 8 Relative degradation under the same partial-occlusion protocol on three corpora (curves are means, shading is ±1 standard deviation; all six models cover the ten seeds 11–139). Left: GitHub issues (background to summary length ratio ≈ 12.9); middle: civil engineering corpus (≈ 11.2); right: AG News (≈ 3.9). Only on GitHub is the degradation of FC-NTM below all baselines; on the civil corpus and AG News the smallest degradation belongs to BTM and LDA respectively, and the gap to FC-NTM is within 0.003–0.013.

FC-NTM on all three corpora (0.486 and 0.438 versus 0.423 on AG News). The field view thus serves as a descriptive read-out available to any model with a topic–word matrix.

5.8 Ablation: fusion reconstruction and gating

The three-step ablation separates two questions: whether the fused mixture receives its own reconstruction term, and whether fusion is averaged or gated. The fusion-reconstruction term is effective, but only for the read-out that uses the fused posterior: adding it under mean fusion raises the class TF–IDF C_V and NPMI by 0.014 and 0.013 (3/3 seeds) while lowering the β read-out by 0.013 and 0.021. Replacing mean fusion by a gate brings no further gain in this corpus and protocol (class TF–IDF differences -0.001 and -0.000, 1/3 seeds; β read-out -0.007 and -0.012, 0/3): within this corpus and protocol, the gate adds no gain over mean fusion once the fusion-reconstruction term is present.

5.9 Ablation: core components

Four components are switched off one at a time. First, removing the JS consistency term causes no consistent change in topic quality on any corpus or read-out (all six cells $|\Delta C_V| \leq 0.004$, none with a clear positive paired majority): in this parameterisation the fused representation is trained by the fusion-reconstruction objective, not by pulling the two posteriors together. Second, the observed effect of cross-field reconstruction and word masking depends on the corpus: on GitHub, removing cross-field reconstruction even raises the β read-out by 0.021 (2/3 seeds) and costs only 0.005 in the class TF–IDF read-out, while on AG News and the civil corpus both read-outs drop when either component is removed (cross-field reconstruction $-0.011/-0.022$ and $-0.022/-0.019$; word masking $-0.026/-0.043$ and $-0.010/-0.012$, with all class TF–IDF columns at 0/3). Third, whole-field dropout acts mostly on the class TF–IDF read-out and is consistently beneficial on two of the three corpora only: it lowers both

Table 9 Field-restricted read-out (C_V, mean±standard deviation over seeds 11/23/37). Every topic–word matrix is truncated to the vocabulary of one field and scored against that field's test text only; "background view" is the abstract/body side and "summary view" the title side. FC-NTM's summary view is scored on 20–22 scorable topics per seed on GitHub (29–30 on AG News, 26–28 on the civil corpus) because topics whose ten field words never occur in the held-out text get an undefined C_V.

Corpus	Model	Background view C_V	Summary view C_V
GitHub issues	FC-NTM	0.697±0.016	0.426±0.012
	BTM	**0.710±0.011**	0.396±0.018
	NMF	0.609±0.004	0.433±0.002
	LDA	0.543±0.011	0.466±0.005
	ProdLDA	0.575±0.006	**0.467±0.034**
AG News	FC-NTM	0.566±0.016	0.423±0.015
	BTM	**0.586±0.015**	0.399±0.018
	NMF	0.477±0.015	0.444±0.002
	LDA	0.419±0.007	**0.486±0.006**
	ProdLDA	0.499±0.032	0.438±0.007
Civil corpus	FC-NTM	**0.637±0.015**	0.408±0.005
	BTM	0.589±0.014	0.382±0.008
	NMF	0.600±0.009	0.398±0.013
	LDA	0.554±0.008	0.415±0.006
	ProdLDA	0.576±0.025	**0.419±0.012**

read-outs on GitHub and AG News but changes the β read-out of the civil corpus by only +0.001 while lowering its class TF–IDF read-out by 0.034. Fourth, field asymmetry does not explain the component-level differences: the civil corpus has almost the same length ratio as GitHub (11.2 versus 12.9) yet behaves like AG News (3.9).

5.10 Sensitivity to the fusion weight

At $\lambda_f = 0.25$ the class TF–IDF read-out is essentially the no-fusion result (C_V 0.713 versus 0.714) while the β read-out has already dropped by 0.024; between 0.5 and 1.0 the class TF–IDF gain is stable (0.727 versus 0.723, NPMI 0.274 versus 0.277, differences below 0.005), and the β penalty narrows from -0.020 to -0.005 relative to $\lambda_f = 0$. The value 0.5 used throughout is therefore a conservative compromise.

5.11 Whole-field occlusion

Deleting the background field entirely leaves topic-word coherence almost unchanged (all six cells within ±0.02, with sign flips across seeds on GitHub), so coherence is insensitive to field loss and the information loss must be read from the document-level posterior and the word-level prediction: about 66% (GitHub) and 48% (AG News) of the documents change their most probable topic, and the per-token NLL rises by 0.24/0.23 with dropout training (0.24/0.27 without). The influence of whole-field

Table 10 Three-step fusion ablation on GitHub issues ($K = 30$, seeds 11/23/37, same protocol). Within one read-out each row changes exactly one thing: first the fused mixture θ_{fuse} is decoded into the reconstruction term (mean fusion), then mean fusion is replaced by a learnable gate (the main model). "Positive pairs" counts the seeds whose paired difference against the row above is positive (C_V, NPMI).

Read-out	Configuration	C_V	NPMI	Positive pairs	UMass
β	No fusion	0.721±0.016	0.266±0.012	—	−2.63
	Mean fusion + fusion recon.	0.708±0.016	0.245±0.013	1/3, 0/3	−2.98
	Gated fusion (main model)	0.701±0.016	0.233±0.012	0/3, 0/3	−3.16
cTFIDF	No fusion	0.714±0.001	0.262±0.007	—	−1.86
	Mean fusion + fusion recon.	0.728±0.007	0.275±0.015	3/3, 3/3	−1.91
	Gated fusion (main model)	0.727±0.022	0.274±0.026	1/3, 1/3	−1.90

Table 11 Core-component ablation on three corpora and two read-outs ($K = 30$, seeds 11/23/37, 700-epoch budget, all other settings identical to the main model). Each column switches off one component: cross-field reconstruction of both streams, the JS consistency weight, field-level word masking, and whole-field dropout; the fusion-reconstruction term is kept in all columns. Values are mean paired ΔC_V (negative means the component matters), parentheses count positive pairs.

Corpus	Read-out	−Cross-field recon.	−JS consistency	−Word masking	−Field dropout
GitHub	β	+0.021 (2/3)	−0.001 (1/3)	−0.012 (1/3)	−0.023 (1/3)
GitHub	Class TF-IDF	−0.005 (1/3)	+0.004 (2/3)	−0.004 (1/3)	−0.021 (1/3)
AG News	β	−0.011 (2/3)	+0.003 (1/3)	−0.026 (0/3)	−0.010 (1/3)
AG News	Class TF-IDF	−0.022 (0/3)	−0.002 (1/3)	−0.043 (0/3)	−0.023 (1/3)
Civil corpus	β	−0.022 (0/3)	+0.001 (2/3)	−0.010 (1/3)	+0.001 (1/3)
Civil corpus	Class TF-IDF	−0.019 (0/3)	−0.002 (0/3)	−0.012 (0/3)	−0.034 (0/3)

dropout is concentrated in the posterior distribution and is weaker for word-prediction calibration. It lowers the per-token increase from 0.269 to 0.232 and the posterior JS from 0.060 to 0.049 on AG News, is directionally the same on the civil corpus (0.125→0.103, JS 0.039→0.033), and shows no increase difference on GitHub (0.242 versus 0.241) where only the JS slightly improves. Posterior drift under field loss is reduced on two of the three corpora. We read these measures next to absolute prediction accuracy because they describe different aspects of the same result.

5.12 Concatenated-field single-encoder control

To compare the full method with a single-encoder alternative under the same budget we train a control on the concatenated bag of words $[x_{\text{bg}}; x_{\text{sm}}]$ with the same shared linear decoder, field weights, KL warm-up and orthogonality penalty; whole-field dropout is emulated by zeroing the background half of the input, and there is no fusion or consistency term. The concatenated input doubles the first encoder layer,

Table 12 Sensitivity to the fusion-reconstruction weight λ_f on GitHub issues (gated fusion, seeds 11/23/37). $\lambda_f = 0$ is the no-fusion setting, $\lambda_f = 0.5$ is the value used throughout; all other settings match the main model.

λ_f	β read-out		Class TF–IDF	
	C_V	NPMI	C_V	NPMI
0	0.721±0.016	0.266±0.012	0.714±0.001	0.262±0.007
0.25	0.697±0.015	0.217±0.031	0.713±0.015	0.262±0.004
0.5	0.701±0.016	0.233±0.012	0.727±0.022	0.274±0.026
1.0	0.716±0.012	0.246±0.026	0.723±0.012	0.277±0.018

Table 13 Whole-field occlusion at inference time ($K = 30$, final configuration, 700-epoch budget). ΔC_V is the coherence difference between full-field and occluded inference under the same read-out; ΔNLL is the per-token increase; agreement is the fraction of documents whose most probable topic is unchanged. The two rows of a corpus use the same seeds, isolating the effect of the training-time dropout setting.

Corpus	Training setting	C_V^{full}	C_V^{occ}	ΔC_V	ΔNLL(per token)	Agreement	Mean JS
GitHub	Dropout ($p = 0.5$)	0.727	0.717	+0.009	+0.241	0.345	0.082
	No dropout	0.705	0.725	−0.020	+0.242	0.346	0.092
Civil corpus	Dropout ($p = 0.5$)	0.636	0.627	+0.009	+0.103	0.673	0.033
	No dropout	0.602	0.597	+0.005	+0.125	0.642	0.039
AG News	Dropout ($p = 0.5$)	0.612	0.594	+0.018	+0.232	0.520	0.049
	No dropout	0.589	0.582	+0.007	+0.269	0.498	0.060

keeping the parameter count comparable (a 1.5% difference), and both models use the same budget, vocabulary and read-outs on all three corpora (seeds 11/23/37). Table 15 reports the comparison.

Across the six corpus–read-out cells, the mean C_V differences between FC-NTM and the control range from +0.010 to +0.029, with paired majorities from 1/3 to 3/3. The control also supports field-restricted read-out and title-only inference.

5.13 Title-only routing on the civil corpus

When the abstract of a record is missing, can its title still be routed to an interpretable civil subfield? The evaluation labels are the OpenAlex primary topics of each record OpenAlex (2026). OpenAlex assigns these labels automatically from the title, abstract and source metadata; the frozen snapshot (topic IDs, fetch date) is released with the code, and the label set is the 15 most frequent topics of the training split, covering 42.9% of the test records (2000-record test split). Every model infers a topic distribution from the title only (FC-NTM and the control use their missing-background inference path); the per-model topic-to-label mapping is a supervised step — the majority training label of each argmax topic, with the training-label prior as fallback

Table 14 Title-only routing to the 15 most frequent OpenAlex topics of the civil corpus (seeds 11/23/37; 857 test records, coverage 42.9%). The supervised TF–IDF classifier provides a reference point on the same label set.

Method	Input	Accuracy	Macro-F1
TF–IDF logistic regression (supervised)	title	0.810	0.805
FC-NTM	title + abstract	0.719±0.016	0.676±0.032
FC-NTM	title only	0.664±0.004	0.624±0.013
Concatenated control	title only	0.653±0.012	0.614±0.034
BTM	title only	0.600±0.013	0.529±0.043
LDA	title only	0.532±0.039	0.458±0.057

Table 15 Concatenated-field single-encoder control (seeds 11/23/37; same budget, vocabulary and read-outs). Δ is FC-NTM minus the control, with positive pairs in parentheses.

Corpus	Read-out	FC-NTM	Concatenated	Δ (positive pairs)
GitHub	β	0.701±0.016	0.683±0.014	+0.017 (2/3)
GitHub	Class TF–IDF	0.727±0.022	0.710±0.020	+0.017 (1/3)
AG News	β	0.586±0.026	0.574±0.019	+0.012 (2/3)
AG News	Class TF–IDF	0.612±0.014	0.584±0.009	+0.028 (3/3)
Civil corpus	β	0.633±0.014	0.623±0.014	+0.010 (2/3)
Civil corpus	Class TF–IDF	0.636±0.014	0.606±0.014	+0.029 (2/3)

for model topics that receive no training record (three test records in one run triggered this fallback) — fitted on the same title-only inference, so the pipeline combines an unsupervised representation with a supervised training-split mapping. Accuracy and macro-F1 are reported per seed. A supervised TF–IDF logistic regression on the same titles provides a reference point, and FC-NTM given both fields marks the upper end.

Title-based routing reaches 0.664 accuracy for FC-NTM, 0.653 for the concatenated control, 0.719 with the full fields and 0.810 for the supervised TF–IDF classifier. The corresponding accuracies for the classical topic models are 0.532 for LDA and 0.600 for BTM. Accuracy and macro-F1 for each seed are reported in Table 14.

6 Discussion and limitations

The results assign the field-aware representation a practical role: one trained model supports field-restricted topic words and keeps inferring topic proportions when the background field is missing. Multi-seed read-out differences are largest and most consistent on the civil corpus and are supported for the matched class TF–IDF read-out on GitHub. AG News marks the boundary: its class TF–IDF mean difference is -0.009 and its confidence interval includes zero. Field cooperation is thus not a blanket win; it pays off on particular corpora, and only for particular read-outs.

The concatenated-field encoder stays close to FC-NTM, so the results characterise the complete field-aware configuration rather than two-stream modelling in general. Relative degradation, absolute NLL and topic agreement capture different aspects of missing-field behaviour, while redundancy and field-length statistics describe corpus structure.

The routing experiment uses automatically generated OpenAlex labels for a fixed 15-topic subset covering 42.9% of the test records. It is useful for screening; deployment will require validation with independent, expert-labelled data.

Scope is the main limitation: the evidence covers three English title/body-style corpora, with ten seeds for FC-NTM and BTM and three for the remaining baselines, and FC-NTM hyper-parameters were selected on development data. Two design couplings bound the ablation evidence: the component ablation attenuates rather than eliminates cross-field information flow, because the fused reconstruction term continues to rebuild both fields; and the gating comparison evaluates the gate together with the fusion-reconstruction term, since the gate receives gradients only through that term. The evidence base is otherwise open on several sides: other languages, other field conventions and automatically detected field boundaries have not been tested, nor has validation on larger independently labelled collections. Within that scope, the released model, protocol and per-seed results give a concrete starting point for missing-field topic modelling on engineering collections.

7 Conclusions

This paper presented FC-NTM, a field-cooperative neural topic model for dual-field short texts: two conditional variational streams share one linear topic decoder, cross-field reconstruction and a consistency regulariser encode the field relation into a shared topic space, and field-level word masking and whole-field dropout handle partial and complete field absence. A single trained model supports a decoder β read-out and a class TF–IDF read-out under one evaluation vocabulary, which removes the vocabulary and topic-length confounds of cross-model comparison.

Across civil engineering title–abstract records and two public corpora, the study identifies the settings in which each read-out produces measurable differences. The largest and most consistent civil-corpus differences are $+0.049$ C_V for class TF–IDF and $+0.061$ for native β (10/10 seeds). Under the matched class TF–IDF read-out, the GitHub difference is $+0.028$ C_V (95% CI $[+0.012, +0.043]$, 9/10). A same-budget concatenated-field encoder provides a close reference, with mean differences of $+0.010$–$+0.029$ and mixed paired majorities. Title-only routing reaches 0.664 accuracy on the evaluated 15-topic subset. The field-relation measurements and occlusion diagnostics separate changes in topic coherence from changes in posterior structure and word-level prediction. The protocol, the trained model and the per-seed results are released together, so the framework can be tried on new engineering collections and checked against the same numbers.

Data availability

The code, the fixed corpus splits and the per-seed experimental results that support this study are publicly available in an online repository (the repository link is provided on the separate title page for double-anonymized review). Trained model checkpoints and the per-seed detail dumps are released as assets of the same repository, and access to the third-party source corpora remains subject to the terms and licences of the original providers. The supplementary material (auxiliary experiments and controls, data and metric notes, example topics and case studies) is included in the same online repository as the code and the per-seed results, under `paper/applied_intelligence/` (the repository link is provided on the separate title page for double-anonymized review).

Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large-language-model assistant (OpenAI Codex with a DeepSeek application programming interface) in order to edit and restructure passages that the authors had drafted, to check numerical consistency between the text, tables and stored result files, and to draft a schematic figure and plotting script that the authors verified against the implementation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The tool was not used to generate research data, to run or select experiments, to produce results, or to draw scientific conclusions, and no figure was generated by a generative image model.

Author contributions

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References

Angelov D (2020) Top2Vec: Distributed representations of topics. ArXiv:2008.09470

Bianchi F, Terragni S, Hovy D (2021) Pre-training is a hot topic: Contextualized document embeddings improve topic coherence. In: Proc., 59th Annual Meeting of the

Association for Computational Linguistics and 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP 2021). Association for Computational Linguistics, Stroudsburg, Pa., pp 759–766, https://doi.org/10.18653/v1/2021.acl-short.96

Bischof JM, Airoldi EM (2012) Summarizing topical content with word frequency and exclusivity. In: Proc., 29th International Conference on Machine Learning (ICML 2012). Omnipress, Madison, Wis.

Blei DM, Ng AY, Jordan MI (2003) Latent Dirichlet allocation. J Mach Learn Res 3:993–1022

Bouma G (2009) Normalized (pointwise) mutual information in collocation extraction. In: Proc., Biennial GSCL Conference 2009 (GSCL 2009). Narr Francke Attempto Verlag, Tübingen, Germany, pp 31–40

Card D, Tan C, Smith NA (2018) Neural models for documents with metadata. In: Proc., 56th Annual Meeting of the Association for Computational Linguistics (ACL 2018). Association for Computational Linguistics, Stroudsburg, Pa., pp 2031–2040, https://doi.org/10.18653/v1/P18-1189

Dieng AB, Ruiz FJR, Blei DM (2020) Topic modeling in embedding spaces. Trans Assoc Comput Linguist 8:439–453. https://doi.org/10.1162/tacl_a_00325

Esmaeili B, Huang H, Wallace BC, et al (2019) Structured neural topic models for reviews. In: Proc., 22nd International Conference on Artificial Intelligence and Statistics (AISTATS 2019). PMLR, pp 3429–3439

Feng J, Zhang Z, Ding C, et al (2022) Context reinforced neural topic modeling over short texts. Inf Sci 607:79–91. https://doi.org/10.1016/j.ins.2022.05.098

García-Méndez S, de Arriba-Pérez F, Barros-Vila A, et al (2023) Automatic detection of relevant information, predictions and forecasts in financial news through topic modelling with Latent Dirichlet Allocation. Applied Intelligence 53(16):19610–19628. https://doi.org/10.1007/s10489-023-04452-4

Grootendorst M (2022) BERTopic: Neural topic modeling with a class-based TF-IDF procedure. ArXiv:2203.05794

Kingma DP, Ba J (2015) Adam: A method for stochastic optimization. In: Proc., 3rd International Conference on Learning Representations (ICLR 2015). OpenReview.net

Kurle R, Günnemann S, van der Smagt P (2019) Multi-source neural variational inference. In: Proc., 33rd AAAI Conference on Artificial Intelligence (AAAI 2019). AAAI Press, Palo Alto, Calif., pp 4114–4121, https://doi.org/10.1609/aaai.v33i01.33014114

Lau JH, Newman D, Baldwin T (2014) Machine reading tea leaves: Automatically evaluating topic coherence and topic model quality. In: Proc., 14th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2014). Association for Computational Linguistics, Stroudsburg, Pa., pp 530–539, https://doi.org/10.3115/v1/E14-1056

Lee DD, Seung HS (2001) Algorithms for non-negative matrix factorization. In: Proc., 14th Annual Conference on Neural Information Processing Systems (NIPS 2000), Advances in Neural Information Processing Systems, vol 13. MIT Press, Cambridge, Mass., pp 556–562

Li X, Zhang A, Li C, et al (2019) Relational biterm topic model: Short-text topic modeling using word embeddings. Comput J 62(3):359–372. https://doi.org/10.1093/comjnl/bxy037

Liu D, Fang Z, Wu B (2026) DVI-DTM: Dual-view representation learning for interpretable short text dynamic topic modeling. In: Proc., 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026). Association for Computational Linguistics, Stroudsburg, Pa., pp 28576–28593, https://doi.org/10.18653/v1/2026.acl-long.1318

Liu W, Huang Y, Guo Y, et al (2022) Topic modeling for short texts via dual view collaborate optimization. In: Proc., 2022 7th IEEE International Conference on Data Science in Cyberspace (DSC 2022). IEEE, Piscataway, N.J., pp 160–166, https://doi.org/10.1109/DSC55868.2022.00028

Miao Y, Yu L, Blunsom P (2016) Neural variational inference for text processing. In: Proc., 33rd International Conference on Machine Learning (ICML 2016). PMLR, pp 1727–1736

Mimno D, Wallach H, Talley E, et al (2011) Optimizing semantic coherence in topic models. In: Proc., 2011 Conference on Empirical Methods in Natural Language Processing (EMNLP 2011). Association for Computational Linguistics, Stroudsburg, Pa., pp 262–272

Moon S, Chi S, Im SB (2022) Automated detection of contractual risk clauses from construction specifications using bidirectional encoder representations from transformers (BERT). Autom Constr 142:104465. https://doi.org/10.1016/j.autcon.2022.104465

Newman D, Lau JH, Grieser K, et al (2010) Automatic evaluation of topic coherence. In: Proc., 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT 2010). Association for Computational Linguistics, Stroudsburg, Pa., pp 100–108

Nguyen T, Doan DT, Luong HT, et al (2026) Sub-document neural topic model. Knowl-Based Syst 331:114762. https://doi.org/10.1016/j.knosys.2025.114762

OpenAlex (2026) Topics. URL https://help.openalex.org/data/topics/, last updated Oct. 7, 2026

OpenIndex (2026) OpenGitHub Issues. URL https://huggingface.co/datasets/open-index/open-github-issues

Pergola G, Gui L, He Y (2021) A disentangled adversarial neural topic model for separating opinions from plots in user reviews. In: Proc., 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2021). Association for Computational Linguistics, Stroudsburg, Pa., pp 2870–2883, https://doi.org/10.18653/v1/2021.naacl-main.228

Pham HTTL, Han S (2023) Natural language processing with multitask classification for semantic prediction of risk-handling actions in construction contracts. J Comput Civ Eng 37(6):04023027. https://doi.org/10.1061/JCCEE5.CPENG-5218

Prechelt L (1998) Early stopping—but when? In: Neural Networks: Tricks of the Trade, Lecture Notes in Computer Science, vol 1524. Springer, Berlin, Germany, p 55–69, https://doi.org/10.1007/3-540-49430-8_3

Priem J, Piwowar H, Orr R (2022) OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. In: Proc., 26th International Conference on Science and Technology Indicators (STI 2022). University of Granada, Granada, Spain, p sti22193, https://doi.org/10.5281/zenodo.6936227

Qin S, Wu H, Zhou L, et al (2024) TAE: Topic-aware encoder for large-scale multi-label text classification. Applied Intelligence 54(8):6269–6284. https://doi.org/10.1007/s10489-024-05485-z

Reimers N, Gurevych I (2019) Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In: Proc., 2019 Conference on Empirical Methods in Natural Language Processing and 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP 2019). Association for Computational Linguistics, Stroudsburg, Pa., pp 3982–3992, https://doi.org/10.18653/v1/D19-1410

Roberts ME, Stewart BM, Tingley D (2019) stm: An R package for structural topic models. J Stat Softw 91(2):1–40. https://doi.org/10.18637/jss.v091.i02

Röder M, Both A, Hinneburg A (2015) Exploring the space of topic coherence measures. In: Proc., 8th ACM International Conference on Web Search and Data Mining (WSDM 2015). ACM, New York, pp 399–408, https://doi.org/10.1145/2684822.2685324

Salama DM, El-Gohary N (2016) Semantic text classification for supporting automated compliance checking in construction. J Comput Civ Eng 30(1):04014106. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000301

Shamshiri A, Ryu KR, Park JY (2024) Text mining and natural language processing in construction. Autom Constr 158:105200. https://doi.org/10.1016/j.autcon.2023.105200

Srivastava A, Sutton C (2017) Autoencoding variational inference for topic models. In: Proc., 5th International Conference on Learning Representations (ICLR 2017). OpenReview.net

Srivastava N, Hinton G, Krizhevsky A, et al (2014) Dropout: A simple way to prevent neural networks from overfitting. J Mach Learn Res 15(56):1929–1958

Suzuki M, Nakayama K, Matsuo Y (2016) Joint multimodal learning with deep generative models. ArXiv:1611.01891

Tixier AJP, Hallowell MR, Rajagopalan B, et al (2016) Automated content analysis for construction safety: A natural language processing system to extract precursors and outcomes from unstructured injury reports. Autom Constr 62:45–56. https://doi.org/10.1016/j.autcon.2015.11.001

Vangelova S, Forth K, Beetz J, et al (2026) Computational taxonomy development for digital passports using semantic topic modelling to advance circular construction. Autom Constr 192:107220. https://doi.org/10.1016/j.autcon.2026.107220

Vorontsov K, Frei O, Apishev M, et al (2015) BigARTM: Open source library for regularized multimodal topic modeling of large collections. In: Proc., Analysis of Images, Social Networks and Texts (AIST 2015), Communications in Computer and Information Science, vol 542. Springer, Cham, Switzerland, pp 370–381, https://doi.org/10.1007/978-3-319-26123-2_36

Wang Y, Taylor JE (2019) DUET: Data-driven approach based on latent Dirichlet allocation topic modeling. J Comput Civ Eng 33(3):04019023. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000819

Webber W, Moffat A, Zobel J (2010) A similarity measure for indefinite rankings. ACM Trans Inf Syst 28(4):1–38. https://doi.org/10.1145/1852102.1852106

Wu C, Li X, Guo Y, et al (2022) Natural language processing for smart construction: Current status and future directions. Autom Constr 134:104059. https://doi.org/10.1016/j.autcon.2021.104059

Wu X, Li C, Zhu Y, et al (2020) Short text topic modeling with topic distribution quantization and negative sampling decoder. In: Proc., 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP 2020). Association for Computational Linguistics, Stroudsburg, Pa., pp 1772–1782, https://doi.org/10.18653/v1/2020.emnlp-main.138

Wu X, Nguyen T, Luu AT (2024) A survey on neural topic models: Methods, applications, and challenges. Artif Intell Rev 57(2):18. https://doi.org/10.1007/s10462-023-10661-7

Yan X, Guo J, Lan Y, et al (2013) A biterm topic model for short texts. In: Proc., 22nd International World Wide Web Conference (WWW 2013). ACM, New York, pp 1445–1456, https://doi.org/10.1145/2488388.2488514

Yin J, Wang J (2014) A Dirichlet multinomial mixture model-based approach for short text clustering. In: Proc., 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2014). ACM, New York, pp 233–242, https://doi.org/10.1145/2623330.2623715

Zhang F, Fleyeh H, Wang X, et al (2019) Construction site accident analysis using text mining and natural language processing techniques. Autom Constr 99:238–248. https://doi.org/10.1016/j.autcon.2018.12.016

Zhang G, Iwata T, Kashima H (2017) Robust multi-view topic modeling by incorporating detecting anomalies. In: Proc., European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2017), Lecture Notes in Computer Science, vol 10535. Springer, Cham, Switzerland, pp 238–250, https://doi.org/10.1007/978-3-319-71246-8_15

Zhang K, Zhou Y, Chen Z, et al (2022) Incorporating biterm correlation knowledge into topic modeling for short texts. Comput J 65(3):537–553. https://doi.org/10.1093/comjnl/bxaa079

Zhang X, Zhao J, LeCun Y (2015) Character-level convolutional networks for text classification. In: Proc., 29th Annual Conference on Neural Information Processing Systems (NIPS 2015), Advances in Neural Information Processing Systems, vol 28. Curran Associates, Red Hook, N.Y., pp 649–657

Zhou P, El-Gohary N (2016) Ontology-based multilabel text classification of construction regulatory documents. J Comput Civ Eng 30(4):04015058. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000530

Zhou P, Qi Y, Yang Q, et al (2025) Neural topic modeling of machine learning applications in building: Key topics, algorithms, and evolution patterns. Autom Constr 170:105890. https://doi.org/10.1016/j.autcon.2024.105890

Zhu B, Cai Y, Zhang H (2021) Sparse biterm topic model for short texts. In: Proc., Web and Big Data: 5th International Joint Conference (APWeb-WAIM 2021), Lecture Notes in Computer Science, vol 12858. Springer, Cham, Switzerland, pp 227–241, https://doi.org/10.1007/978-3-030-85896-4_19

Zhu Q, Feng Z, Li X (2018) GraphBTM: Graph enhanced autoencoded variational inference for biterm topic model. In: Proc., 2018 Conference on Empirical

Methods in Natural Language Processing (EMNLP 2018). Association for Computational Linguistics, Stroudsburg, Pa., pp 4663–4672, https://doi.org/10.18653/v1/D18-1495